

Quiz 2 (40 points)

Due: 20/11, Wed., in class.

Questions are testing: modes of convergence. You might need some techs from Borel-Cantell lemma, and convergence theorems.

1. Prove the statement or disprove it by providing at least one counter example: Convergence in L^p imply almost surely.
2. State the definition of uniform integrability (UI).
3. Proof the statement: A family $\{X_t : t \in T\}$ of integrable random variables defined on some probability space $(\Omega, \mathcal{F}, \mathbb{P})$ is uniformly integrable (UI) if and only if the following two statements hold true.
 - (i) The family X_t is bounded in L^1 .
 - (ii) For any $\varepsilon > 0$, there exists $\delta > 0$ such that

$$\text{for any } F \in \mathcal{F}, \mathbb{P}(F) < \delta \text{ then } \mathbb{E}[|X_t| 1_F] < \varepsilon \quad \text{for all } t \in T.$$

Proof. ($\implies$) Since the family $\{X_t : t \in \mathcal{T}\}$ is UI. Given any $\varepsilon > 0$, there exists $\lambda > 0$ such that

$$\mathbb{E}[|X_t| 1_{\{|X_t| > \lambda\}}] \leq \varepsilon \quad \text{for any } t \in \mathcal{T}.$$

Then we can have

$$\mathbb{E}[|X_t|] = \mathbb{E}[|X_t| 1_{\{|X_t| \leq \lambda\}}] + \mathbb{E}[|X_t| 1_{\{|X_t| > \lambda\}}] \leq \lambda + \varepsilon,$$

Now for the above λ and any $F \in \mathcal{F}$, we can get

$$\begin{aligned} \mathbb{E}[|X_t| 1_F] &= \mathbb{E}[|X_t| 1_{\{|X_t| \leq \lambda\} \cap F}] + \mathbb{E}[|X_t| 1_{\{|X_t| > \lambda\} \cap F}] \\ &\leq \lambda \mathbb{P}(F) + \mathbb{E}[|X_t| 1_{\{|X_t| > \lambda\}}] < \lambda \mathbb{P}(F) + \varepsilon. \end{aligned}$$

Take $\delta \triangleq \varepsilon/\lambda$. For $\mathbb{P}(F) < \delta$, it holds

$$\mathbb{E}[|X_t| 1_F] < 2\varepsilon \quad \text{for any } t \in \mathcal{T}.$$

This gives the uniform continuity property in (ii).

($\impliedby$) Since X_t is bounded in L^1 , by Markov's inequality, we can get

$$\mathbb{P}(|X_t| > \lambda) \leq \frac{1}{\lambda} \mathbb{E}[|X_t|] \leq \frac{1}{\lambda} M \quad \text{for any } t \in \mathcal{T}.$$

Therefore, given $\varepsilon > 0$ there exists $\lambda > 0$ such that

$$\hat{\lambda} > \lambda \implies \mathbb{P}(|X_t| > \hat{\lambda}) < \delta,$$

then we can obtain

$$\mathbb{E}[|X_t| 1_{\{|X_t| > \hat{\lambda}\}}] < \varepsilon \quad \text{for any } t \in \mathcal{T}.$$

This gives the UI. The proof is completed. $\square$

4. If $X_n \rightarrow X$ in probability, and $\{X_n\}_{n=1}^\infty$ are independent, show that X must be a constant random variable. How about the case $X_n \rightarrow X$ a.s., can we get the limit is a constant random variable?

*** You may try to the proof by contradiction.

Proof. The case of convergence in probability:

Suppose for a contradiction that X is not constant, so that there are some $a < b$ and ε such that $\mathbb{P}(X < a) \geq \varepsilon$ and $\mathbb{P}(X > b) \geq \varepsilon$.

Convergence in probability implies convergence in distribution, so eventually we can get the distribution function convergence in the sense that $\mathbb{P}(X_n < a) \geq \varepsilon/2$ and the same for $\mathbb{P}(X_n > b) \geq \varepsilon/2$ for some large n . Therefore with probability $(\varepsilon/2)^2 > 0$, for m, n large we have that $\{X_n < a\}$ and $\{X_m > b\}$ happen at the same time.

On the other hand, convergence in probability implies for $n, m \rightarrow \infty$, let $\delta = (b - a)/3$, we have the probability of events $|X_n - X| \leq \delta$ and $|X_m - X| \leq \delta$ is also 1. This means there is a positive probability that $\{X_n < a\} \cap \{|X_n - X| \leq \delta\}$ and $\{X_m > b\} \cap \{|X_m - X| \leq \delta\}$ occur, so it is impossible.

It is also possible to use that there is a subsequence that converges almost surely, but infinitely often we have $X_n < a$ and also infinitely often $X_n > b$.

The case of convergence almost surely:

By Kolmogorov 0-1 law, we let $F(X) = \mathbb{P}(X \leq x)$. Since $\{X \leq x\} = \{X \leq x\} \cap \{X \leq x\}$ we get $F(X) = F(x)F(x)$. Hence $F(x) = 0$ or 1 for each x . If $c = \sup\{x : F(x) = 0\}$ then $F(x) = 0$ for $x < c$ and 1 for $x \geq c$. This implies that $\mathbb{P}(X \neq c) = 0$ so $X = c$ almost surely. $\square$